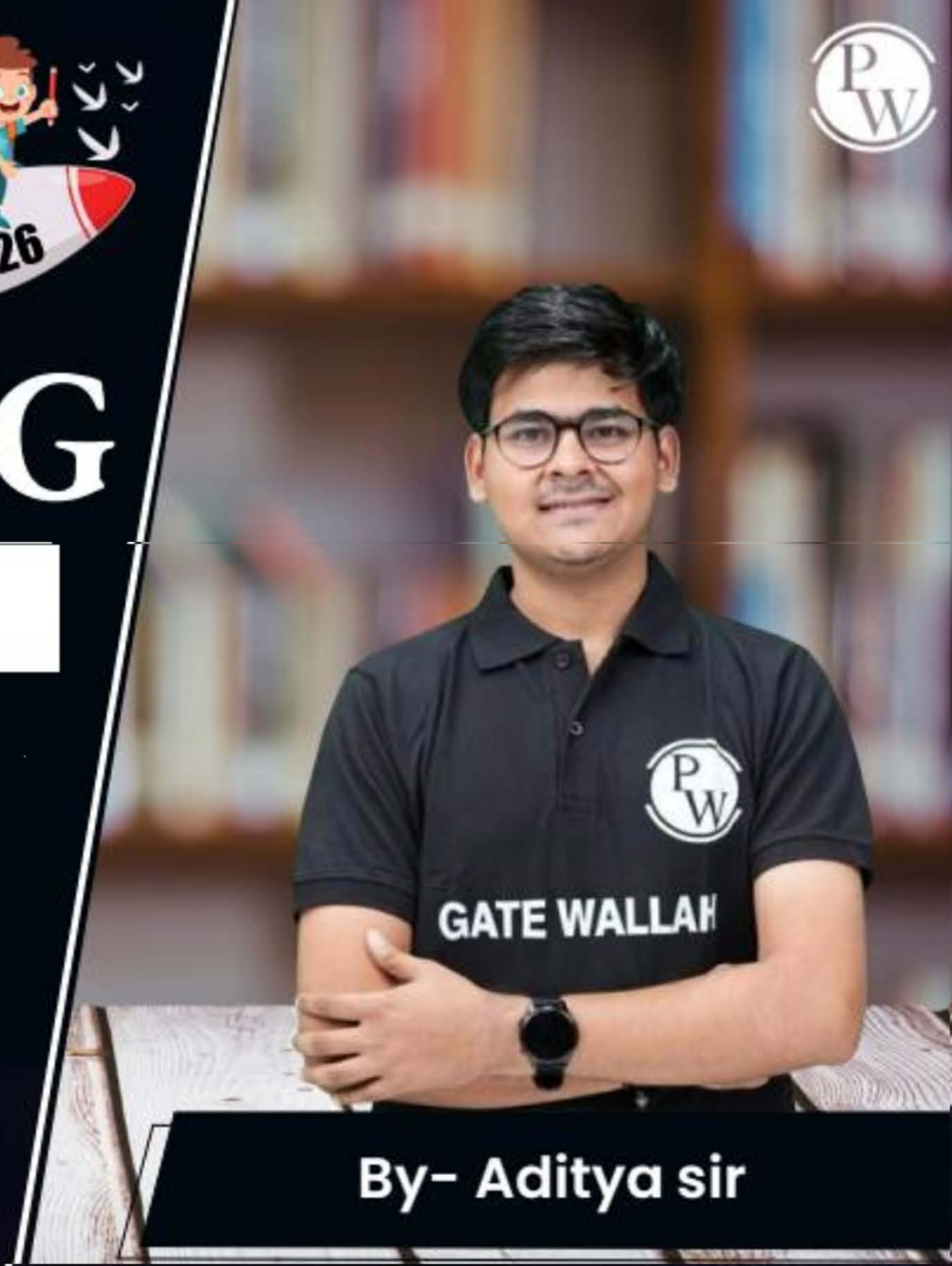
CS & IT ENGINEERING



Algorithms

Greedy Method

Lecture No.- 03



By- Aditya sir

Recap of Previous Lecture



Topic

Topic

Fractional Knapsack

Imp Terminologies

Job Seq with Deadline (JSD)

Topics to be Covered



- Topic
- Topic
- Topic

JSD

Optimal Merge Patterns



About Aditya Jain sir

1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on Linkedin where I share my insights and guide students and professionals.



Telegram Link for Aditya Jain sir: https://t.me/AdityaSir_PW

Enumeration based JSD:-

$|J| = 0$

$|J| = 1$

$|J| = 2$

:

} we are checking all possible subsets

→ 2^n subsets



Topic : Greedy of Algorithms

- Job sequencing with deadline (JSD):

1. Enumeration logic:

   → ~~$O(n^2)$~~ $O(2^n)$

Hence,

Need greedy based Algo to reduce down the TC.



Topic : Greedy Algorithms

Engineers mindset

Eg.1. n = 4

Jobs	(Di) Deadline	(Pi) Profit	
J1	2	200	.....(1)
J2	1	30	.....(4)
J3	2	50	.....(3)
J4	1	80	.....(2)

Enumeration

N→ $O(2^n)$

↓

Expo

Max(D) =2

Max profit =280



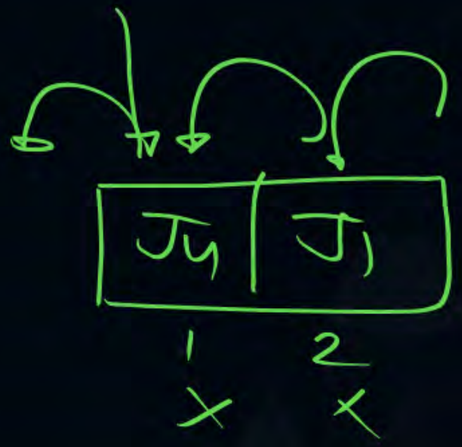
Topic : Greedy Algorithms

Greedy based optimal solution for previous Ques.

Logic:-

1. Max (Deadline) → Queue size of

J4	J1
1	2



2. Sort jobs on basis of decreasing profit

J1 → 2 → 200 (✓)

J4 → 1 → 80 (✓)

J3 → 2 → 50 (✗)

J2 → 1 → 30 (✗)

Max profit greedy appr.

= 200 + 80 + 0 + 0

= 280



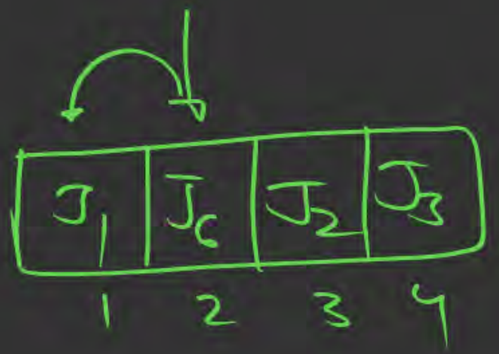
Topic : Greedy Algorithms

#Q.

n=6	(Di) Deadline	(Pi) Profit
J1	2	28
J2	3	36
J3	4	24
J4	2	20
J5	4	16
J6	3	30

3 ✓
1 ✓
4 ✓
5 ×
6 ×
2 ✓

Max(D) = 4



Profit = 118

Topic : Greedy Algorithms

#Q.

n=6	(Di) ~~Benefit~~ Deadline	(Pi) ~~Profit~~
J1	56	2
J2	24	1
J3	10	5
J4	36	1
J5	60	2
J6	40	3

J1 ..... (2)
J2 ..... (5)
J3 ..... (6)
J4 ..... (4)
J5 ..... (1)
J6 ..... (3)



Ans :- 166

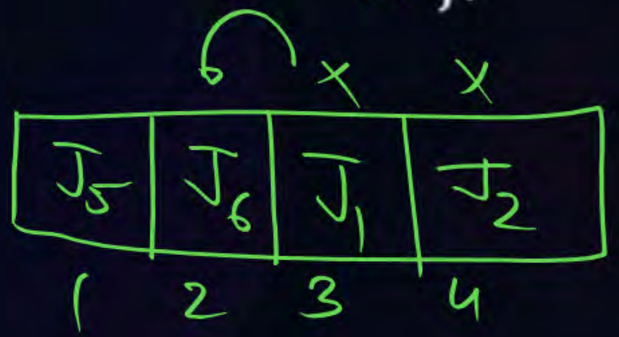
Max(D) = 5

Topic : Greedy Algorithms

#Q.

	Deadline	Profit	
✓ J1	4	140	2)
✓ J2	4	170	1)
× J3	4	24	6)
× J4	4	36	5)
✓ J5	3	100	4)
✓ J6	4	120	3)
× J7	3	20	7)

84!



↳ 4

Ans :- 530

Max(D) = 4

Max profit =?



Topic : Greedy Algorithms

```
1.  Algorithm Greedy Job(d, J, n)
2.  // J is a set of jobs that can be completed by t heir deadlines.
3.  {
4.      J : {1};
5.      For i : = 2 to n do
6.      {
7.          If (all jobs in j U {i} can be completed
8.              By their deadlines) then J: = J U {i}
9       }
10. }
```

Assumption?

→ Jobs are given in descending order of their profits.



Topic : Greedy Algorithms

P4Q

#Q. We are given 9 tasks T1, T2...., T9. The execution of each task requires one unit of time. We can execute one task at a time. Each tasks T1 has a profit Pi and a deadline di, profit pt is earned if the task is completed before the end of the Deadline.

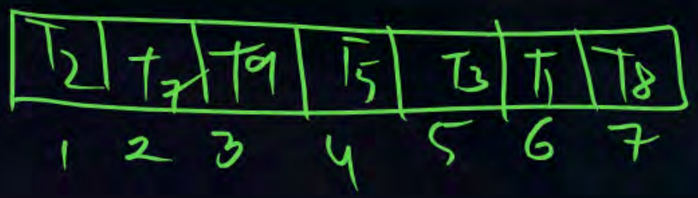
Task	T1	T2	T3	T4	T5	T6	T7	T8	T9
Profit	15	20	30	18	18	10	23	16	25
Deadline	7	2	5	3	4	5	2	7	3

A) ✗ all done

B) ✗ T1 & T3

C) ✗ T1 & T6

D) ✗ T4 & T6





Topic : Greedy Algorithms

#Q. Are all tasks completed in the schedule that gives maximum profit?

A All tasks are completed

B T1 and T3 are left out

C T1 and T6 are left out

D T4 and T6 are left out



Topic : Greedy Algorithms

#Q. What is the maximum profit earned?

96.7%

A 147

B 165

C 167

D 175

Optimal Merge Patterns

(omp)



Topic : Greedy Algorithms

P3: Optimal merge patterns:- (Application of merging algo.)

Given ‘n’ files

It is reqd to merge them using 2-way merging, such that the total number of record moments are minimum.

Default:- [Merging 2-files at a time]



Topic : Greedy Algorithms

Concept:- V Imp

Given 2 sorted files 'A' & 'B' having 'n' and 'm' records respectively. To merge them into a single sorted file, it requires (n + m) record movements.

(n+m-1) max Comparisons

The task is to minimize the Total no. of Record movements.



Topic : Greedy Algorithms

Eg.2.

n =3

A: [2, 3, 5]

m = 4

B : [1, 4, 8, 10]

[1, 2, 3, 4, 5, 8, 10,]

Record movement = n+ m

= 3 + 4

= 7

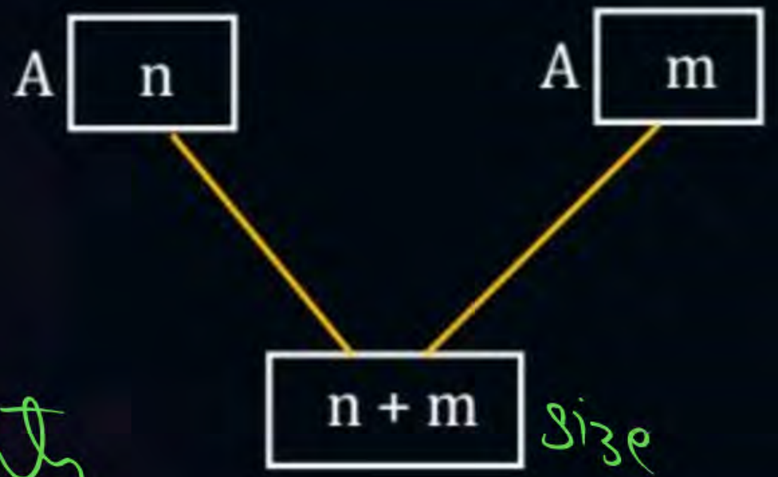


Topic : Greedy Algorithms

2-way merging Algorithm (merge sort)

Number of elements comparisons:

V. Imp



(n+m) Record movements

Best case = min (n,m)

Worst case= (m + n –1) element comparisons.



eg:- A = 10 = no. of Records in file 'A'

B = 15

C = 5

30X







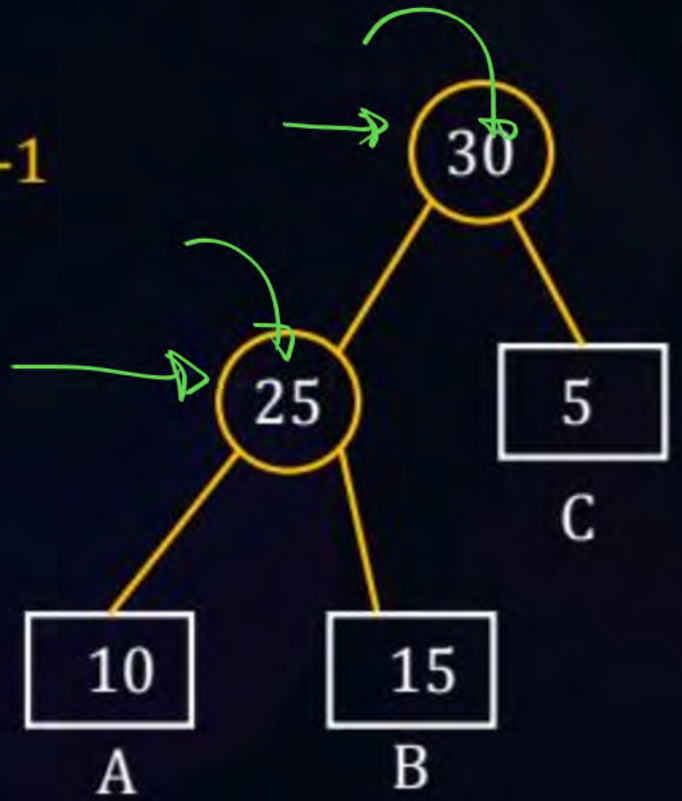
Topic : Greedy Algorithms

Eg.3. A: 10 → "number of records in files A"

B: 15 → "number of records in files B"

C: 5 → "number of records in files C"

Case-1



Total no. of records movement = 25 + 30
= 55

Order = ABC

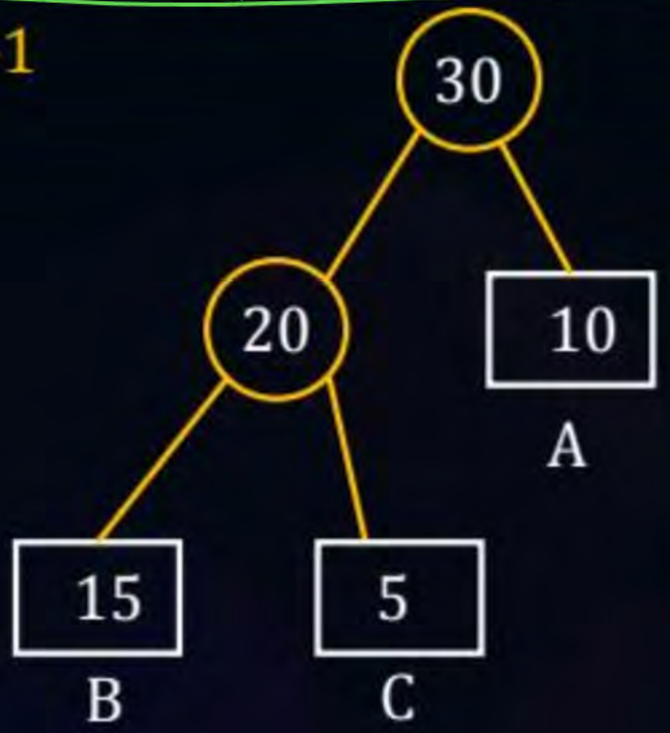


Topic : Greedy Algorithms

optimal

2 - Way binary merge tree

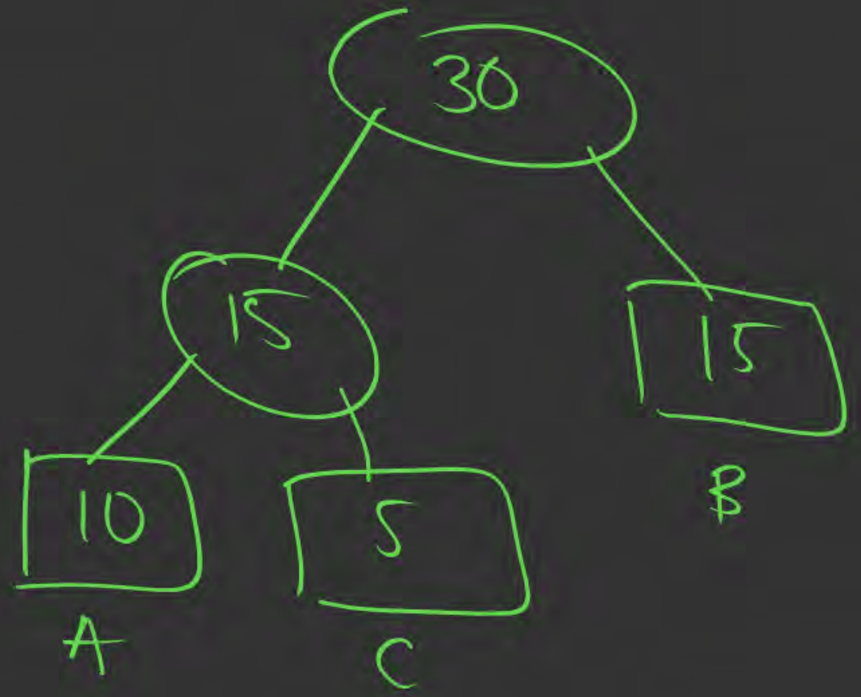
Case-1



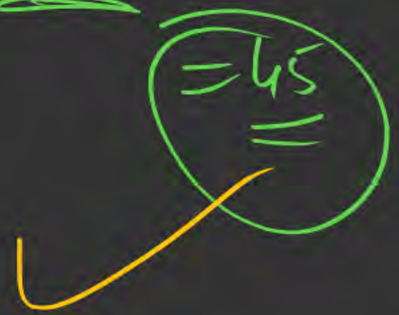
Total no. of records movement = 20 + 30
= 50

Order = BCA

3) AC B



Total RM = 15 + 30





Topic : Greedy Algorithms

Observation.

1. Ordering matters and impacts the total number of records movements.
2. For ‘n’ files, how many times 2-way merging operation is required ? ⇒(n-1)
3. Objective:- Minimize the total number of records movements.



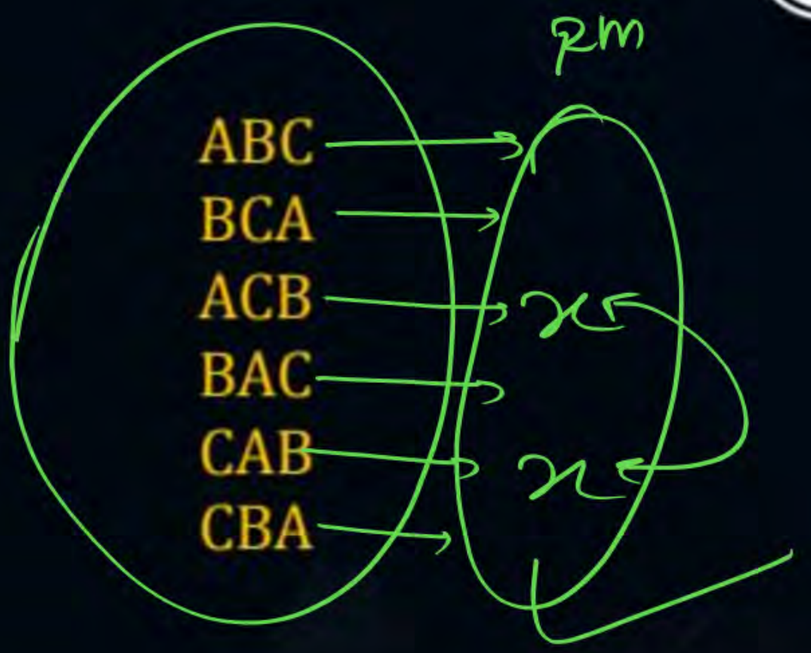
Topic : Greedy Algorithms

Observation.

4. Size of solution space = ?

n! arrangement

All n! Solution are feasible.



Optimal Soln is always unique ⇒ (45)



Topic : Greedy Algorithms



Appr.-1

Brute force/ Enumeration

guarantees ~~Queerants~~ optimal sol.

- But time complexity = $O(n!)$

$= O(n^n)$

Hence, we need a based solution → greedy algo.

Greedy based optimal Soln.

(OMP)

Topic : Greedy Algorithms

max comp = RM − (n−1)

Appr.-2

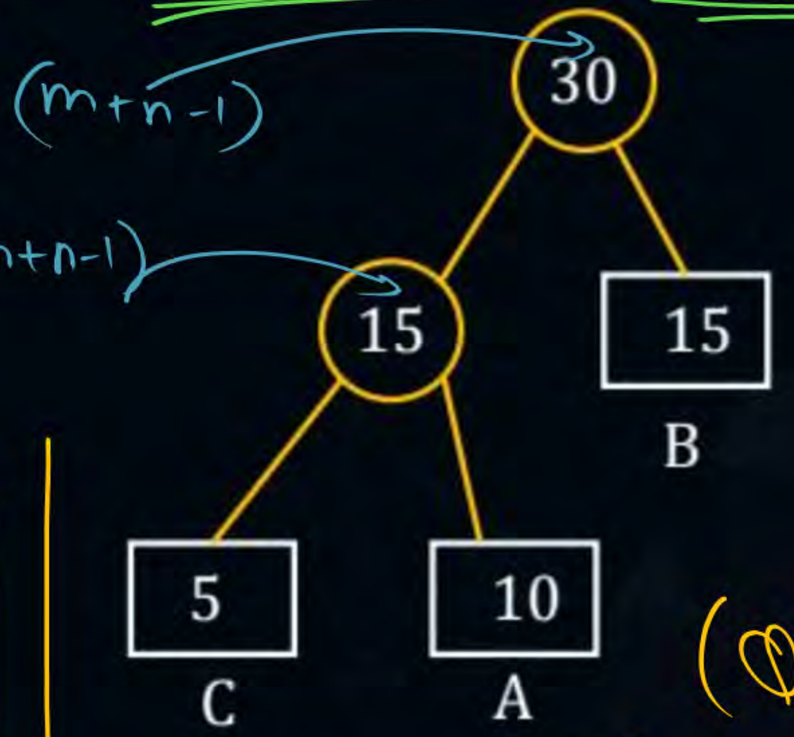
Greedy algorithm

E.g. A : 10 — 2

B : 15 — 3

C : 5 — 1

(Small → left Larger → right)



Total number of record movement
= 15 + 30
= 45 → optimal

Total elements – wise comparison
= (15 −1) + (30 −1)
= 14 + 29
= 43



Topic : Greedy Algorithms

#Q. F1 → 12

F2 → 5

F3 → 18

F4 → 10

F5 → 2

HW

Min Record Movements ?



Topic : Greedy Algorithms

Minimum number of records movement required = ?

Total records movement = 7 + 17 + 29 + 47

= 24 + 29 + 47

= 53 + 47

= 100





Topic : Greedy Algorithms

Formally

The total number of record movement = weighted external path length of the 2 – way binary merge tree.

$=\sum_{i=1}^{n} di * q1$

Where

d_i = distance from root to file I

d_i = size/freq. of file



2 mins Summary

- Topic
- Topic
- Topic
- Topic
- Topic

JSD

OPM



THANK - YOU